Problem

For the power triangle in Fig. 11.34(b), the apparent power is:

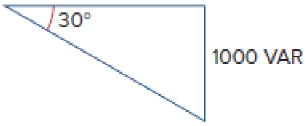
(a) 2000 VA

(b) 1000 VAR

(c) 866 VAR

(d) 500 VAR

Figure 11.34(b):



Step-by-step solution

Step 1 of 2

Refer to Figure 11.34 (b) in the textbook.

Draw the circuit as shown in Figure 1.

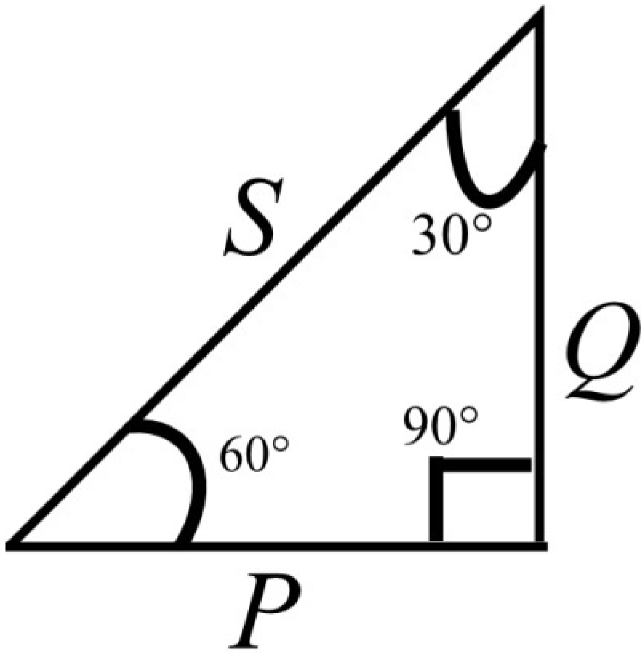


Figure 1

Step 2 of 2

Compare Figure 1 with Figure 11.34 (b).

By comparing,

The real power is $P = 1000\ \text{VAR}$.

Determine the apparent power.

Write the relation between real power and apparent power.

$$P = S\cos(\theta_v - \theta_i)$$

Replace $1000\ \text{VAR}$ for P, $60°$ for $\theta_v - \theta_i$ in equation.

$$1000 = S\cos(60)$$
$$S = \frac{1000}{\cos(60)}$$
$$S = \frac{1000}{0.5}$$
$$S = 2000\ \text{VA}$$

Therefore, the apparent power is $\boxed{2000\ \text{VA}}$.

Therefore, the correct option is $\boxed{(a)}$.